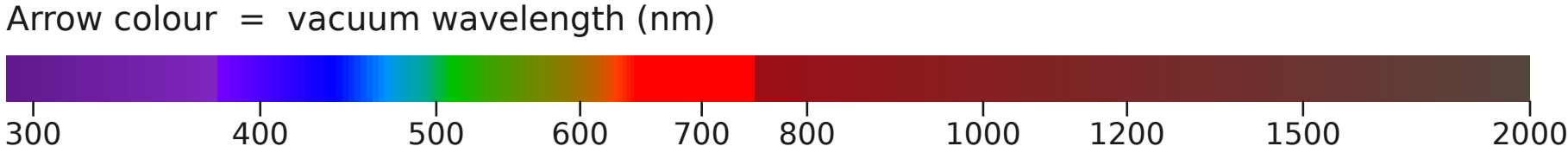


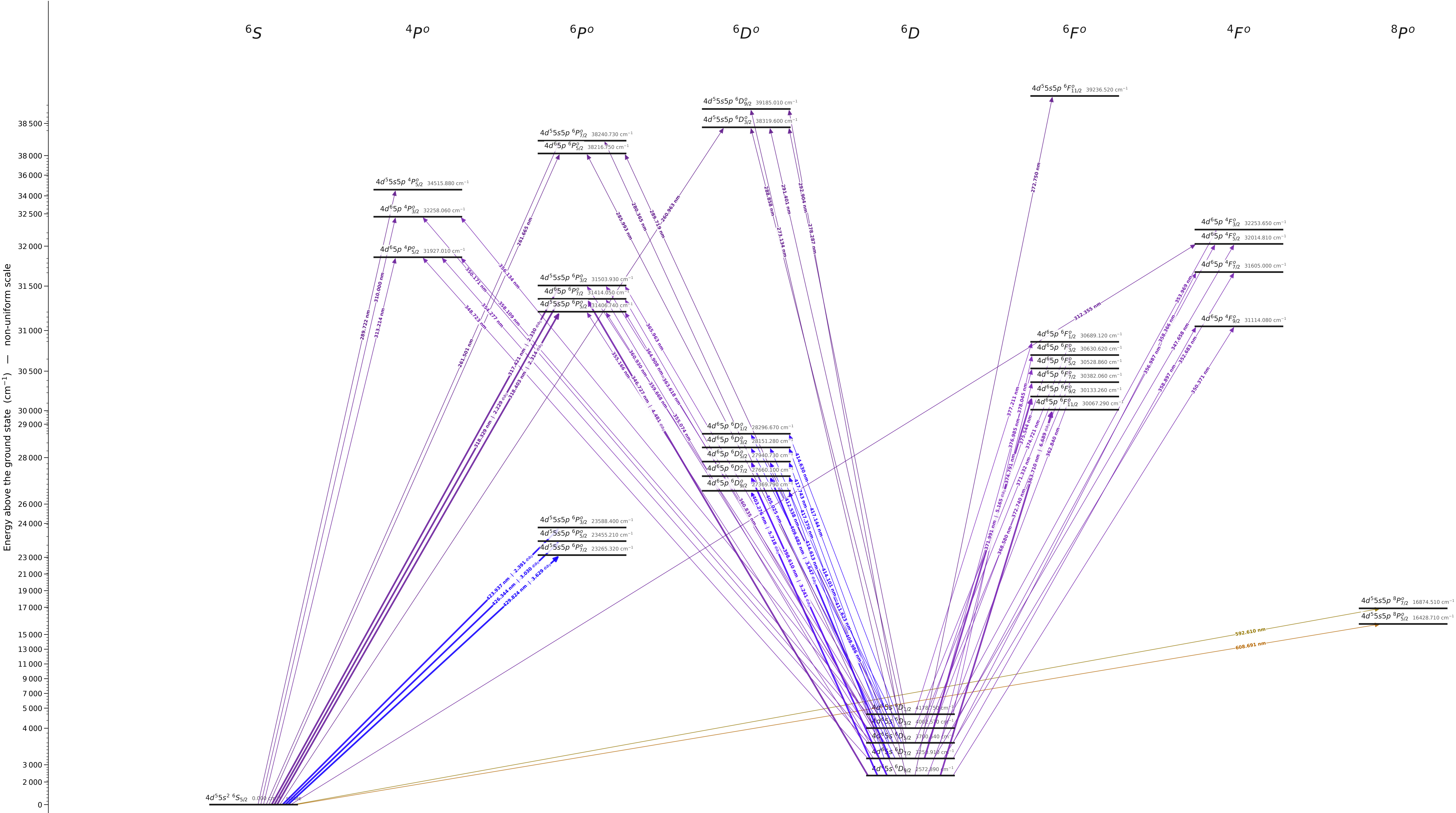
Tc

Arrow label: vacuum wavelength | reduced dipole matrix element $\langle U || e r || U' \rangle$ in ea_0
Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: NIST energy and measured lifetime (where one exists).



Tc (Tc I). Strongest classified transitions below 2 μm (NIST): 37 levels, 70 lines.

Tc⁺ ionisation limit 57421.68 cm^{-1} = 7.11938 eV (174.150 nm)



Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$\langle U e r U' \rangle$ (ea_0)	A (s^{-1})	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
$4d^5 5s^2 \ ^6S_{5/2} - 4d^5 5s 5p \ ^6P_{3/2}^o$	317.4207	317.3289	944.4641	2.330	8.600e+07	-	-	
$4d^5 5s^2 \ ^6S_{5/2} - 4d^6 5p \ ^6P_{3/2}^o$	318.3289	318.2369	941.7695	2.229	3.900e+07	-	-	
$4d^5 5s^2 \ ^6S_{5/2} - 4d^5 5s 5p \ ^6P_{5/2}^o$	318.4030	318.3109	941.5504	2.314	5.600e+07	-	-	
$4d^6 5s \ ^6D_{9/2} - 4d^6 5p \ ^6P_{7/2}^o$	346.7267	346.6274	864.6362	4.481	1.220e+08	-	-	
$4d^6 5s \ ^6D_{9/2} - 4d^6 5p \ ^6F_{11/2}^o$	363.7104	363.6068	824.2614	6.689	1.570e+08	-	-	
$4d^6 5s \ ^6D_{7/2} - 4d^6 5p \ ^6F_{9/2}^o$	371.9913	371.8855	805.9126	5.165	1.050e+08	-	-	
$4d^6 5s \ ^6D_{9/2} - 4d^6 5p \ ^6D_{7/2}^o$	398.6095	398.4968	752.0956	3.241	4.200e+07	-	-	
$4d^6 5s \ ^6D_{9/2} - 4d^6 5p \ ^6D_{9/2}^o$	403.2762	403.1623	743.3924	5.718	1.010e+08	-	-	
$4d^6 5s \ ^6D_{7/2} - 4d^6 5p \ ^6D_{9/2}^o$	409.6818	409.5662	731.7691	3.647	4.900e+07	-	-	
$4d^5 5s^2 \ ^6S_{5/2} - 4d^5 5s 5p \ ^6P_{3/2}^o$	423.9372	423.8178	707.1624	2.391	3.800e+07	-	-	
$4d^5 5s^2 \ ^6S_{5/2} - 4d^5 5s 5p \ ^6P_{5/2}^o$	426.3445	426.2245	703.1695	3.030	4.000e+07	-	-	
$4d^5 5s^2 \ ^6S_{5/2} - 4d^5 5s 5p \ ^6P_{7/2}^o$	429.8243	429.7034	697.4768	3.629	4.200e+07	-	-	

λ , ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
 $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.